

OmniTrade Strategy Blueprint

Generated: June 7, 2026

Stock selection, execution, portfolio sizing, and research diagnostics.

This document describes how the current OmniTrade scanner and portfolio engine selects stocks, ranks signals, and allocates portfolio weight. It is a research and decision-support model, not financial advice.

1. High-Level Objective

OmniTrade separates stock selection into two layers:

- Scanner layer: finds stocks that look attractive either as long-term investments or short-term trading candidates.
- Execution and portfolio layer: takes short-term signals and decides whether they are actionable enough to become portfolio positions.

The scanner can show a stock because it has a good score. The portfolio engine is stricter: a stock must be triggered, have positive expectancy after modeled costs, avoid major risk penalties, and have enough sizing edge.

2. Universe Selection

The model starts from a fixed, liquid universe in `config/universe.py`.

Default universe characteristics:

- Mostly large, liquid US and international equities.
- Minimum price filter.
- Minimum average volume filter.
- Minimum market-cap filter.
- Minimum historical data length filter.

The intent is to avoid tiny, illiquid, noisy names where slippage, stale data, and false signals are harder to control.

3. Long-Term Selection Logic

Long-term scoring starts from a neutral base around 45 and adjusts using composite factors.

Current factor groups:

- Trend factor: price above MA200, MA50 above MA200, six-month return.
- Quality factor: profit margins and return on equity.
- Growth factor: revenue growth and earnings growth.

- Valuation factor: trailing PE when available.
- Balance-sheet factor: debt/equity and current ratio.

Default scanner inclusion:

- ``min_long_term_scan_score = 65``
- Recommendation label must be ``Strong Buy`` or ``Buy``.

Interpretation:

- A strong long-term pick should have trend support plus business-quality or growth support.
- A stock can still score well with incomplete fundamentals, but missing data makes the case more dependent on trend.

4. Short-Term Regimes

Short-term scoring is now split into two explicit regimes.

Momentum

Momentum is trend-following. It rewards stocks that are already moving constructively and are not yet overextended.

Typical momentum profile:

- Price above MA20 and MA50.
- Price above or near VWAP.
- 15-minute and 1-hour RSI supportive but not too stretched.
- Volume rising enough to confirm demand.
- For swing trades, MACD and moving-average structure are supportive.

Mean Reversion

Mean reversion is a pullback setup inside a larger constructive trend. It rewards stocks that are temporarily oversold but not structurally broken.

Typical mean-reversion profile:

- Daily trend remains constructive, especially price above MA200.
- MA50 above MA200 helps confirm the larger trend.
- 15-minute or 1-hour RSI is oversold or reset.
- Price is pulling into VWAP, MA20, or MA50.

- Volume is active but not panic-like.

The app now carries regime metadata such as:

- `regime\_label`
- `regime\_scores`
- `ranking\_bucket`
- `rank\_within\_regime`

This makes it possible to compare momentum stocks against other momentum stocks, and mean-reversion stocks against other mean-reversion stocks.

5. Short-Term Horizons

The app scores two short-term horizons:

- 1-2 day trade:
 - Uses 15-minute RSI, 1-hour RSI, VWAP location, and volume spike.
 - Can produce either momentum or mean-reversion setups.
- 5-15 day swing:
 - Uses daily RSI, MA20, MA50, MA200, volume ratio, and MACD.
 - Can produce either momentum continuation or mean-reversion pullback setups.

The primary short-term score is the stronger of the two horizons.

Default scanner inclusion:

- `min\_short\_term\_scan\_score = 55`
- Recommendation label must be `Strong Setup` or `Watchlist`.

6. Strategy_v1 Execution Logic

Strategy_v1 is the execution layer for short-term signals.

Default requirements:

- Signal trade state must be `ENTER NOW`.
- Signal score must be at least `70`.
- Signal must be from a supported short-term family:

- ``short_term_day``
- ``short_term_swing``
- Entry waits for a pullback of ``0.50%`` below the original signal entry.

That means a signal can look good in the scanner but still wait in the execution layer until price reaches the trigger.

7. Ranking and Edge Score

The execution engine ranks candidates with an edge score.

Current default weights:

- Raw signal score: 30%
- Historical expectancy after modeled costs: 40%
- Trigger proximity: 15%
- Recency: 10%
- Lower volatility: 5%

Important behavior:

- Historical expectancy now has a larger default role than raw score.
- If raw score and expectancy strongly disagree, the row gets an explicit conflict flag.
- Ranking remains deterministic by using stable tie-breakers such as ticker, score, expectancy, and timestamp.

8. Transaction Costs

The model now has a simple transaction-cost scaffold.

Parameters:

- ``commission_per_trade``
- ``slippage_bps``
- ``default_trade_notional``

Net expectancy is computed as:

```
net_expectancy_pct = historical_expectancy_pct - estimated_transaction_cost_pct
```

By default, costs are set to zero and the cost filter is disabled, so current behavior remains close to the original app. Once costs are configured, the portfolio engine prefers net expectancy.

9. Reward/Risk Filter

The execution layer can compute a simple reward/risk ratio using:

- Planned entry or trigger price.
- Target price.
- Stop-loss price.

Default RR setting:

- ``min_reward_risk = 2.0``
- ``reward_risk_filter_enabled = False``

When enabled, a triggered signal can be rejected if the target/stop structure does not meet the minimum reward/risk requirement.

10. Time Stop

The engine now includes a time-stop diagnostic scaffold.

Default settings:

- ``time_stop_enabled = False``
- ``time_stop_max_holding_days = 5``
- ``time_stop_min_favorable_move_pct = 1.0``

When enabled, a triggered trade that fails to move in favor after the configured window can be marked as eligible for exit or downgrade.

This is currently diagnostic because the app does not yet have a true broker-fill or live position ledger.

11. Turnover Awareness

The engine estimates per-ticker trading frequency using recent logged signals.

Default settings:

- ``turnover_lookback_days = 90``
- ``high_turnover_annualized_signal_count = 48``
- ``max_turnover_penalty = 0.30``

If a ticker produces too many signals, its raw portfolio weight can be penalized. The purpose is to avoid names that look active but churn too often.

12. Portfolio Eligibility

A stock must pass the execution layer before becoming a portfolio holding.

Portfolio eligibility requires:

- Trigger status is `triggered`.
- Net expectancy is positive when available.
- Position size is positive.
- Execution decision is not rejected.
- Risk penalty and turnover penalty do not reduce raw weight to zero.

Fragile cohorts can still be reduced sharply by the existing risk logic.

13. Research Logging

The strategy now supports append-only diagnostics logging.

Logging is controlled in `config/performance.py`:

- `RESEARCH_LOGGING_ENABLED = False``
- Log output: ``data_store/research_logs/strategy_execution_diagnostics.jsonl``

When enabled, each execution row logs:

- Timestamp and ticker.
- Regime and horizon.
- Raw scores and labels.
- Historical expectancy and net expectancy.
- Edge score.
- Thresholds in effect.
- Estimated costs.
- Reward/risk.
- Turnover metrics.
- Execution decision.
- Rejection reason when applicable.

This JSONL file is meant for offline analysis and future backtesting.

14. Main Tuning Knobs

Primary files:

- `domain/backtest/thresholds.py`
- `config/performance.py`

Core thresholds:

- `min\_short\_term\_scan\_score`
- `min\_long\_term\_scan\_score`
- `min\_execution\_score`
- `execution\_pullback\_pct`

Ranking weights:

- `EDGE\_SCORE\_WEIGHTS.raw\_signal\_score`
- `EDGE\_SCORE\_WEIGHTS.historical\_expectancy`
- `EDGE\_SCORE\_WEIGHTS.trigger\_proximity`
- `EDGE\_SCORE\_WEIGHTS.recency`
- `EDGE\_SCORE\_WEIGHTS.lower\_volatility`

Cost model:

- `COMMISSION\_PER\_TRADE`
- `SLIPPAGE\_BPS`
- `DEFAULT\_TRADE\_NOTIONAL`
- `COST\_FILTER\_ENABLED`

Risk and execution:

- `REWARD\_RISK\_FILTER\_ENABLED`
- `MIN\_REWARD\_RISK`
- `TIME\_STOP\_ENABLED`
- `TIME\_STOP\_MAX\_HOLDING\_DAYS`
- `TIME\_STOP\_MIN\_FAVORABLE\_MOVE\_PCT`

Turnover:

- `TURNOVER\_LOOKBACK\_DAYS`
- `HIGH\_TURNOVER\_ANNUALIZED\_SIGNAL\_COUNT`
- `MAX\_TURNOVER\_PENALTY`

15. Current Strategy Summary

The current default strategy is conservative in execution:

- It scans broadly inside a liquid large-cap universe.
- It separates long-term investing logic from short-term trading logic.
- It now separates short-term momentum from short-term mean reversion.
- It requires `ENTER NOW` plus score ≥ 70 before Strategy_v1 considers execution.
- It waits for a 0.50% pullback.
- It ranks candidates with stronger emphasis on historical expectancy.
- It sizes only positive-expectancy triggered candidates after risk and turnover adjustments.

The main improvement opportunity is to turn the new diagnostics into offline research:

- Compare momentum vs mean-reversion expectancy separately.
- Sweep score thresholds and pullback percentages.
- Test whether RR filtering improves realized returns.
- Test whether cost and turnover penalties reduce churn without removing too much edge.